



Spontaneous particle creation by oscillating compact stars

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Abstract

In quantum field theory, dynamical curved spacetimes can **excite particle pairs from the quantum vacuum** [2]. We study this effect for **radially oscillating compact stars** [3], a strong-gravity setting whose quantum particle-creation problem remains largely unexplored. Using a massless minimally coupled scalar field in the exterior Schwarzschild spacetime, we model the stellar surface as a moving boundary [4] and compute the in/out Bogoliubov coefficients **non-perturbatively**. The resulting spectrum and total particle number show **spontaneous particle creation with a clear resonance structure** [1].

Star Model

By **Birkhoff's theorem**, the exterior of a spherically symmetric star of mass M is **Schwarzschild**:

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad f(r) := 1 - 2\frac{M}{r}.$$

The dynamics enters through the stellar surface, $R_0(t) = R_0 + M\varepsilon(t)\sin(\Omega t)$, with a smooth switch,

$$\varepsilon(t; t_{\text{on}}, t_{\text{off}}, \Delta, A) = \frac{A}{4} \left[1 + \tanh\left(\frac{t - t_{\text{on}}}{\Delta}\right) \right] \left[1 - \tanh\left(\frac{t - t_{\text{off}}}{\Delta}\right) \right].$$

A massless minimally coupled scalar field obeys $g^{ab}\nabla_a\nabla_b\phi(x) = 0$. Spherical symmetry gives independent radial sectors $R_\ell(t, r)Y_{\ell m}(\theta, \phi)$. We model the star-field coupling by the Dirichlet condition $R_\ell(t, R_0(t)) = 0$, treating the stellar surface as a **time-dependent effective mirror**.

The exterior problem is completed by asymptotic flatness: at $r \rightarrow \infty$, we require

$$R_\ell(t, r) \sim \int_{-\infty}^{\infty} d\omega j_\ell(\omega r) e^{-i\omega t}.$$

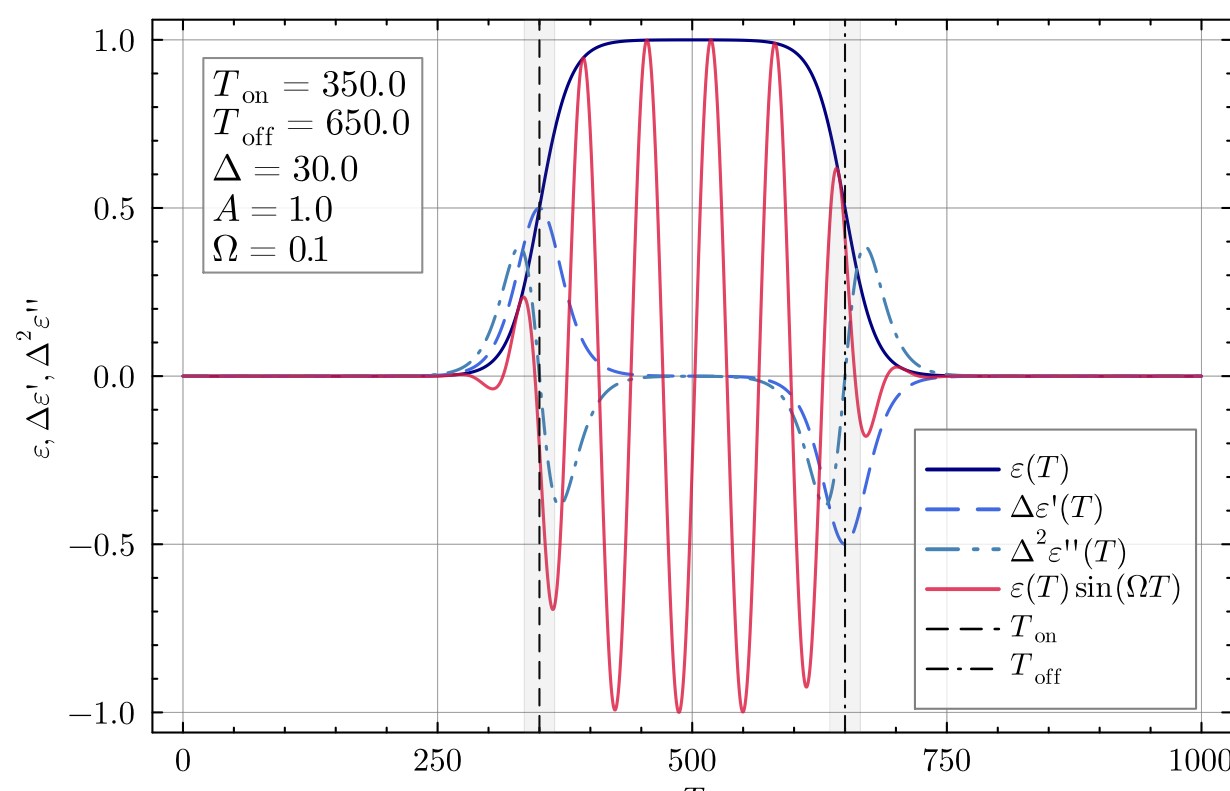


Figure 1: Switching profile and resulting radial oscillation of the stellar surface.

In and out vacua

Since the spacetime is static before t_{on} and after t_{off} , the field operator admits two natural expansions,

$$\hat{\phi}(x) = \int_0^\infty d\omega \sum_{\ell=0}^\infty \sum_{m=-\ell}^\ell \left[a_{\omega\ell m}^{\text{in/out}} \phi_{\omega\ell m}^{\text{in/out}}(x) + a_{\omega\ell m}^{\text{in/out},\dagger} \overline{\phi_{\omega\ell m}^{\text{in/out}}(x)} \right].$$

The corresponding **positive-frequency modes** reduce to stationary modes in the asymptotic regimes,

$$\phi_{\omega\ell m}^{\text{in/out}}(x) = \sqrt{\hbar} A_{\omega\ell} R_{\omega\ell}^{\text{in/out}}(t, r) Y_{\ell m}(\theta, \phi) \approx \sqrt{\hbar} A_{\omega\ell} e^{-i\omega t} R_{\omega\ell}(r) Y_{\ell m}(\theta, \phi),$$

with “in” taken for $t \ll t_{\text{on}}$ and “out” for $t \gg t_{\text{off}}$. These choices define the two vacua,

$$a_{\omega\ell m}^{\text{in}} |\text{in}\rangle = 0, \quad a_{\omega\ell m}^{\text{out}} |\text{out}\rangle = 0.$$

The particle content is determined by the Bogoliubov coefficient

$$\beta_{\vec{k}\vec{k}'} = \frac{i}{\hbar} \int_{R_0(t)}^{r_{\text{max}}} dr r^2 f(r)^{-1} \int_{S^2} d\Omega \left[\phi_k^{\text{in}} \partial_t \phi_{k'}^{\text{out}} - \phi_{k'}^{\text{out}} \partial_t \phi_k^{\text{in}} \right],$$

where $\vec{k} = (\omega, \ell, m)$. After discretizing the radial spectrum at finite r_{max} , the **total out-particle number** becomes

$$\langle \text{in} | N_{\text{out}} | \text{in} \rangle = \frac{\pi^2}{r_{\text{max}}^2} \sum_{\ell=0}^\infty \sum_{n=1}^\infty \sum_{n'=1}^\infty (2\ell+1) |\beta_{n\ell, n'\ell}|^2.$$

Fixed-boundary formulation and determination of initial data

We make the fixed-boundary picture precise through the comoving coordinate transformation

$$T(t, r) = t, \quad z(t, r) = r - S(t) = r - M\varepsilon(t)\sin(\Omega t),$$

so the stellar surface becomes a **fixed** inner boundary. The Klein-Gordon equation becomes a PDE with time-dependent coefficients, with boundary condition $R_\ell(T, R_0) = 0$. In these coordinates, the metric is given by

$$ds^2 = -[f(r) - f(r)^{-1}\dot{S}^2]dT^2 + 2f(r)^{-1}\dot{S}dTdz + f(r)^{-1}dz^2 + r^2d\Omega^2,$$

with $r(T, z) = z + S(T)$. The **radial modes** in the comoving coordinates obey

$$\left[-\partial_T^2 + 2\dot{S}\partial_T\partial_z + (f^2 - \dot{S}^2)\partial_z^2 + \left(2M\frac{f}{r^2} + 2\frac{f^2}{r} + \ddot{S} \right) \partial_z - \ell(\ell+1)\frac{f}{r^2} \right] R_{\omega\ell}(T, z) = 0.$$

which reduces to the static Regge-Wheeler equation whenever $S(T) = 0$. At the outer numerical boundary, we impose a **free-evolution condition**, $R_{\omega\ell}(T, z_{\text{max}}) = e^{-i\omega T} R_{\omega\ell}(z_{\text{max}} + S(T))$, compatible with asymptotic flatness and mode orthogonality; at early and late times it reduces to a fiducial Dirichlet condition at r_{max} .

In the static regimes, we rewrite the radial equation as a **Sturm-Liouville problem** to obtain a discrete, orthogonal basis $R_{n\ell}(z)$ with real frequencies $\omega_{n\ell}$ on the finite numerical domain. This basis normalizes modes, labels particles, and sets clean in/out data:

$$R_{n\ell}^{\text{in}}(T_0, z) = e^{-i\omega_{n\ell}T_0} R_{n\ell}(z), \quad R_{n\ell}^{\text{out}}(T_{\text{max}}, z) = e^{-i\omega_{n\ell}T_{\text{max}}} R_{n\ell}(z).$$

The Dirichlet boundary conditions preserve orthogonality; the continuum is recovered as $r_{\text{max}} \rightarrow \infty$.

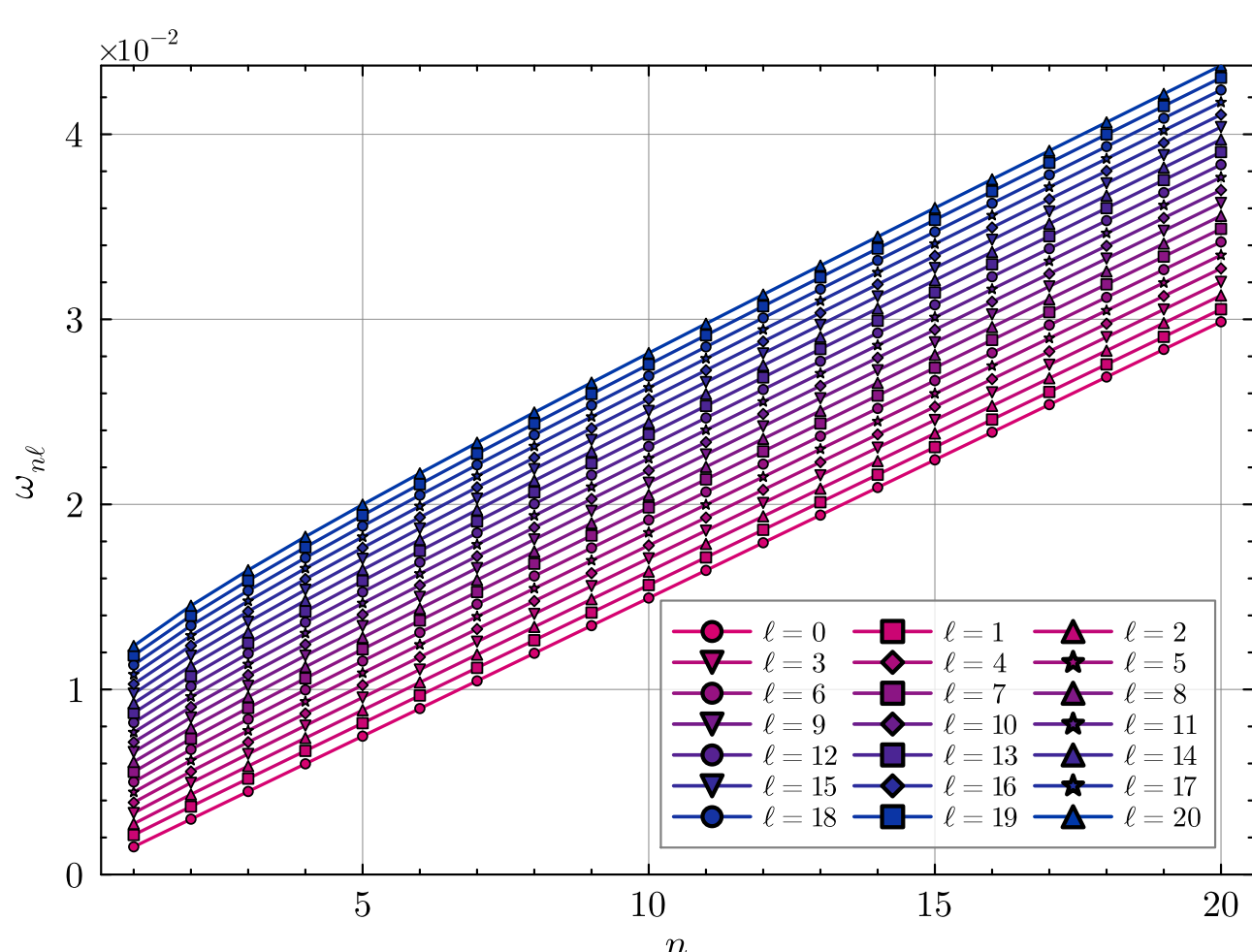


Figure 2: Allowed frequencies $\omega_{n\ell}$.

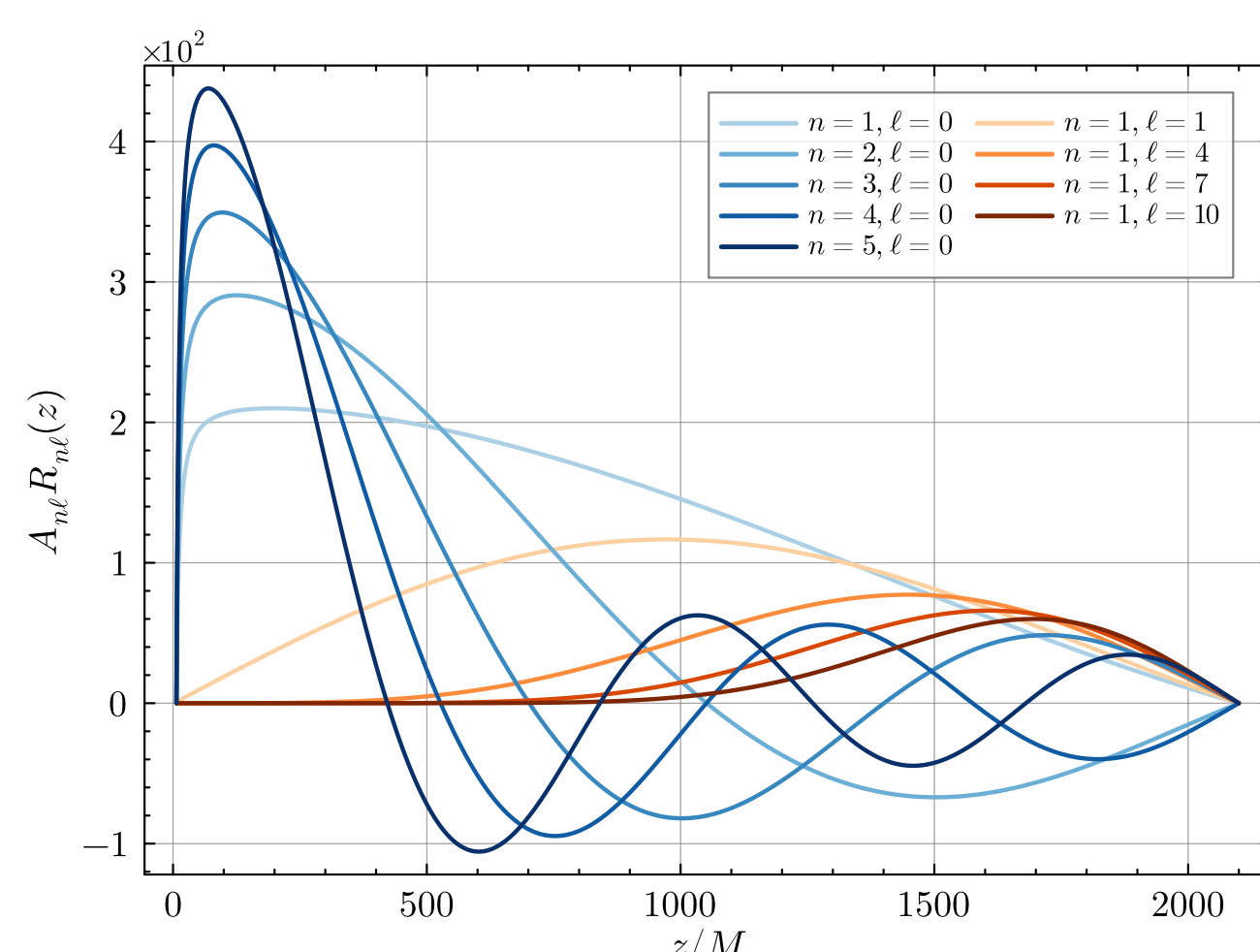


Figure 3: Representative radial eigenfunctions.

Time evolution of the modes

The time-dependent radial PDE is evolved with the **Method of Lines**: finite differences in z and Verner 9/8 adaptive Runge-Kutta in T . We use $C = 2M/R_0 = 0.3$, $R_0 = 20M/3$, $\Omega = 0.03M^{-1}$, $T_{\text{on}} = 765M$, $T_{\text{off}} = 1330M$, and $\Delta = 40M$, with oscillations of 15% of the initial radius.

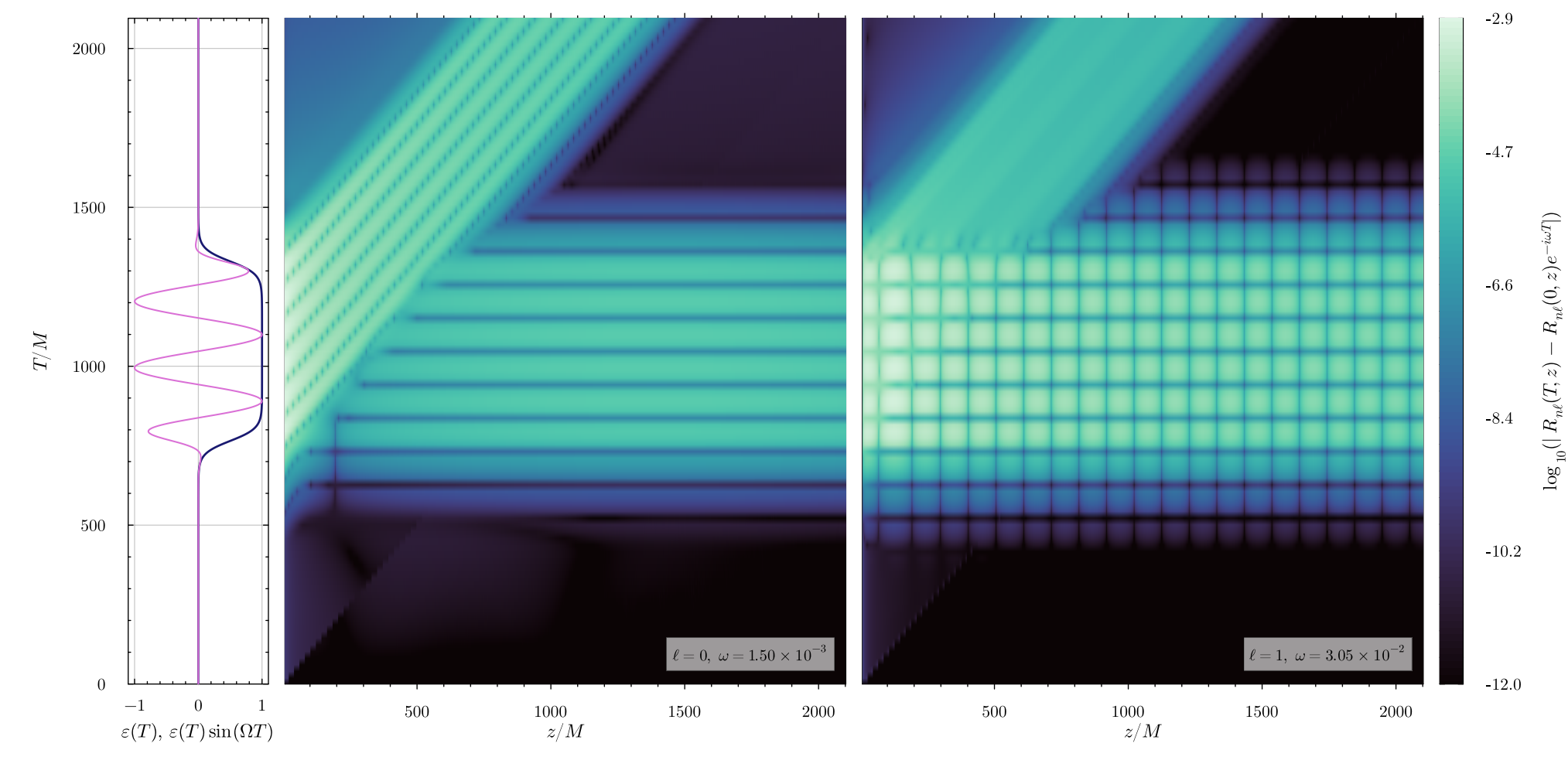


Figure 4: Time evolution for the ground state mode and closest-to-resonance modes.

To keep **boundary artifacts** out of the physical region, the fiducial grid is extended to $2N = 22400$ points. Since disturbances propagate at $c = 1$, reflections from the artificial boundary cannot return during the simulated time window.

The modes show **field perturbations** generated at the stellar surface and propagating outward at the speed of light.

Numerical precision

The **symplectic product** is conserved analytically, giving a direct benchmark: the **Klein-Gordon norm** of every evolved mode remains constant throughout the evolution. If it drifted, β would depend on the hypersurface where it is computed, and the particle spectrum would lose a clear physical meaning.

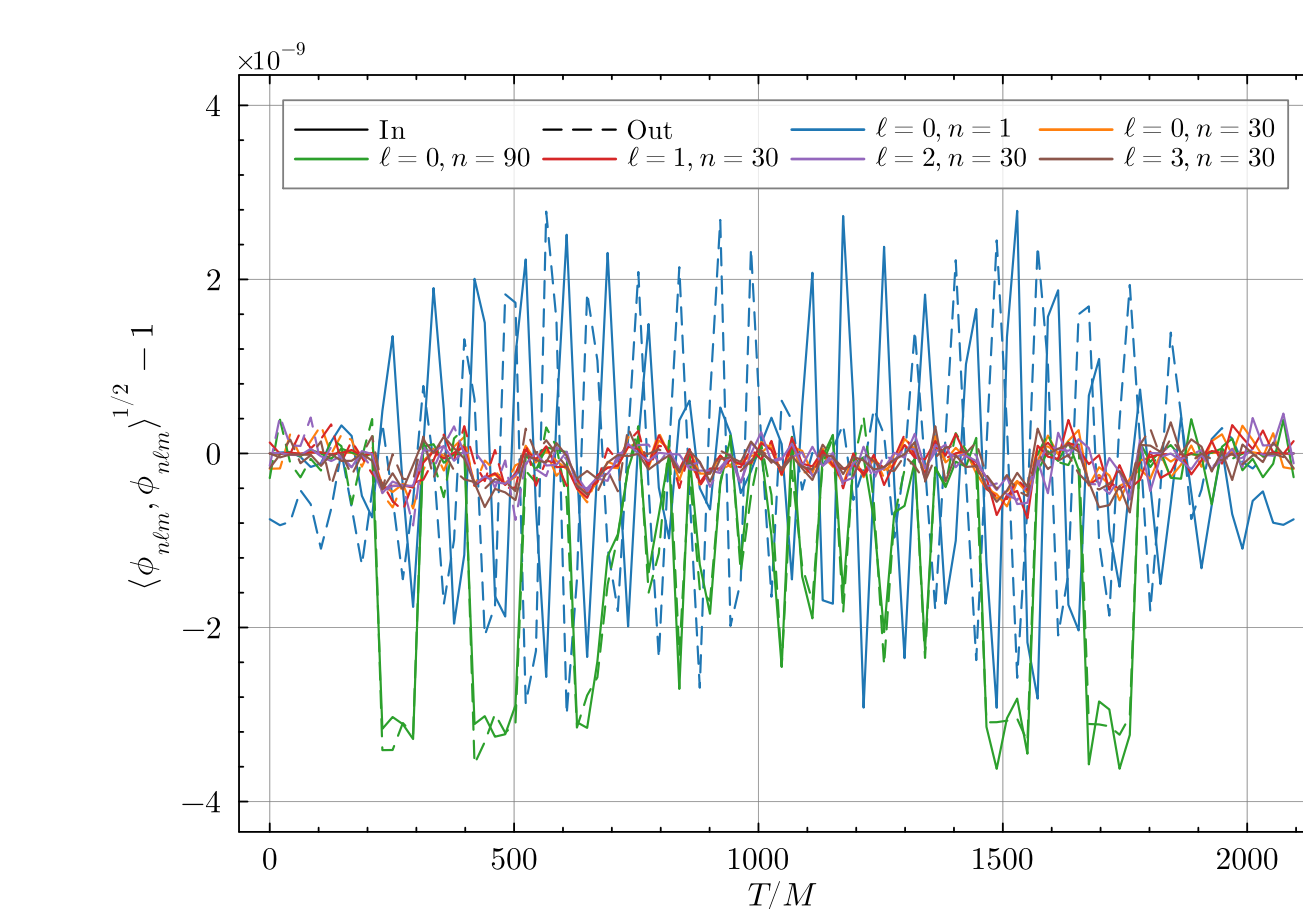


Figure 5: Norm error for representative modes.

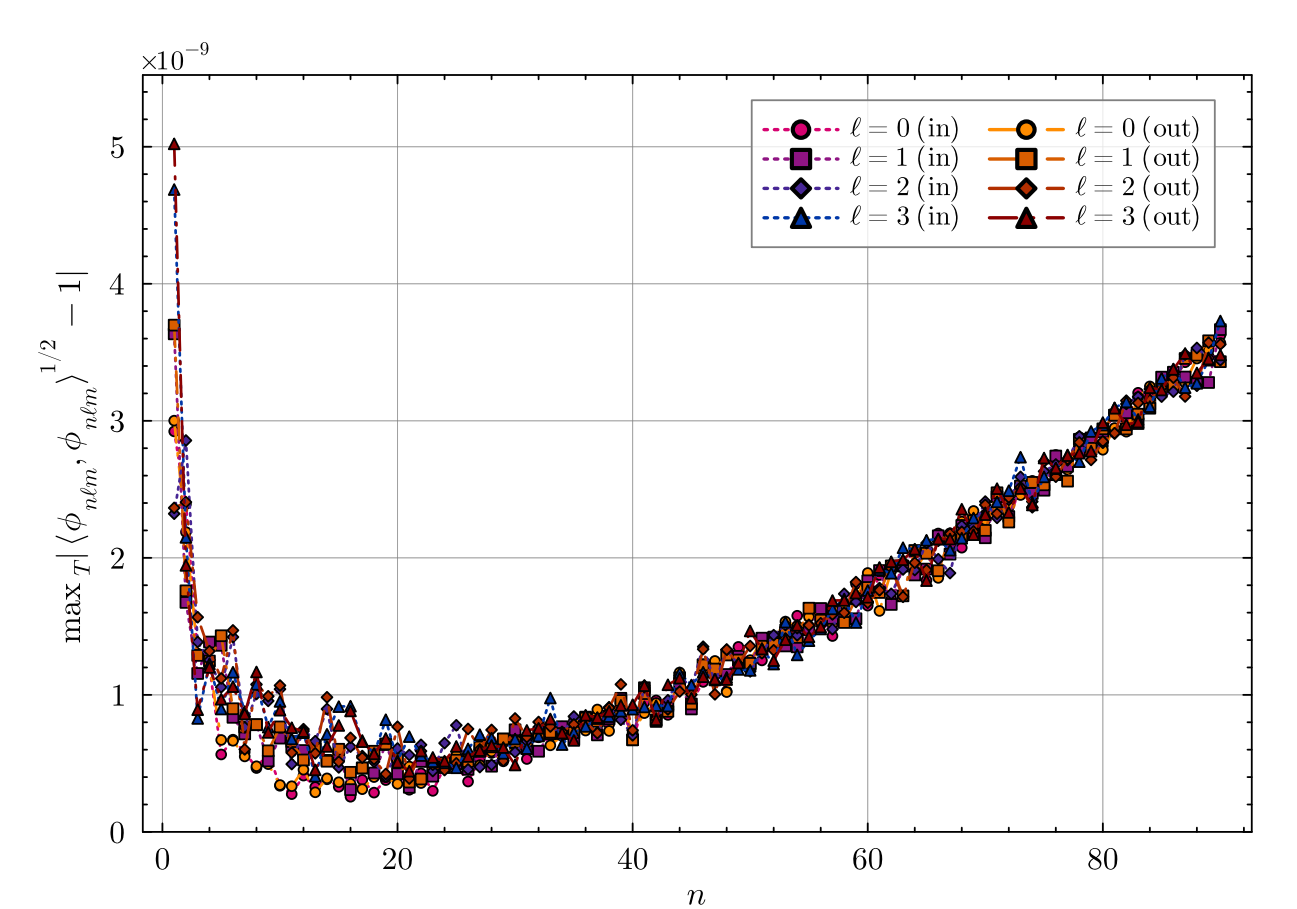


Figure 6: Maximum norm drift vs. n .

Particle Spectrum

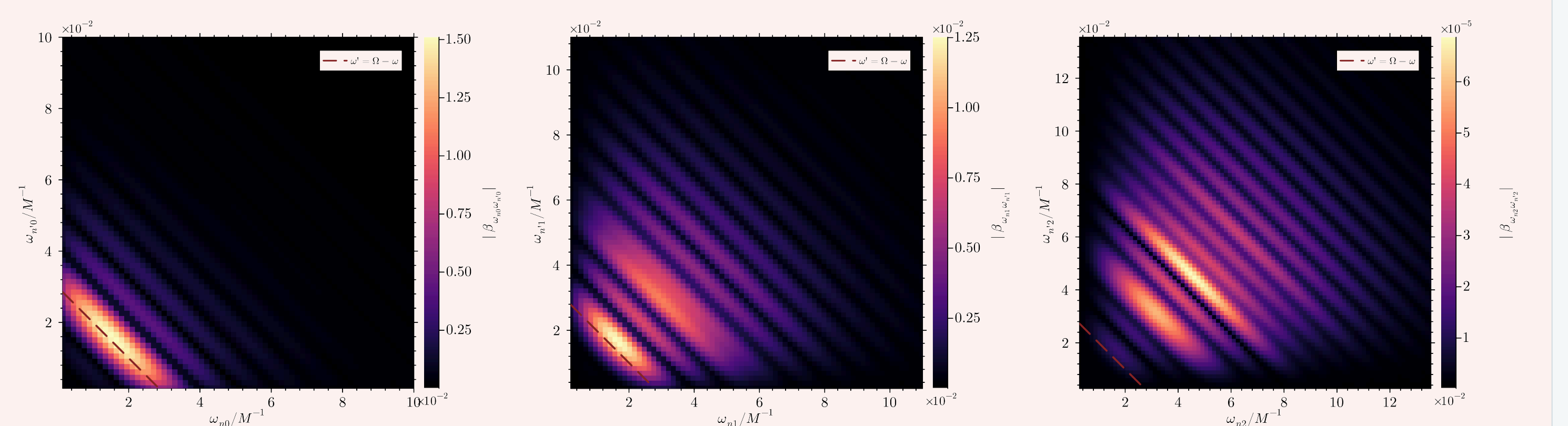


Figure 7: $\ell = 0$.



Figure 8: $\ell = 1$.



Figure 9: $\ell = 2$.

Particle creation is enhanced when the oscillation frequency supplies the combined energy of the produced pair, $\Omega \simeq \omega + \omega'$. The spectra are symmetric under $(n, n') \rightarrow (n', n)$ and decay away from resonance. As $\omega_{n\ell}$ shifts upward with angular momentum, the **resonant band** moves toward smaller accessible mode numbers and eventually becomes suppressed. The β are tied to the **Fourier transform** of the surface acceleration:

$$|\beta_{n\ell, n'\ell}| \propto \left| \mathcal{F} \left\{ \frac{d^2}{dt^2} [M\varepsilon(t)\sin(\Omega t)] \right\} (\omega_{n\ell} + \omega_{n'\ell}) \right|.$$

For $\ell = 0$, this function fits the numerical spectrum to relative deviations of order 10^{-2} . Higher- ℓ sectors receive **graybody-factor corrections** from curvature backscattering.

Total Particle Number

The **total number** is obtained from the spectra above by evaluating $|\beta_{n\ell, n'\ell}|$ at several coordinate times T , averaging the resulting values, and using half the max-min spread as the numerical uncertainty. Since the averaged spectra rapidly decay for frequencies well above Ω , the sums over n and n' can be truncated on the displayed spectral domain. The converged result is dominated by the **s-wave sector**:

$$\sum_{n, n'} |\beta_{n0, n'0}|^2 = (224.344572 \pm 7 \times 10^{-6}) M^2.$$

The $\ell = 3$ sector is already below the uncertainty of the $\ell = 0$ contribution. **Truncating at $\ell \leq 2$** gives

$$\langle \text{in} | N_{\text{out}} | \text{in} \rangle = \pi^2 \frac{M^2}{r_{\text{max}}^2} (224.439475 \pm 7 \times 10^{-6}) \approx 5 \times 10^{-4}.$$

This hierarchy is expected: the perturbation is **purely radial**, so it couples most efficiently to spherically symmetric modes.

Conclusions & Outlook

- Our analysis confirms particle creation by oscillating compact stars in a fully relativistic, non-perturbative framework.
- The created-particle spectra feature a **resonance pattern**. For $\ell = 0$, this follows an analytic estimate tied to the surface acceleration.
- Non-vacuum initial states could add **stimulated emission** without shifting the resonant frequencies [5].
- Natural extensions include **spin-1 fields**, **realistic stellar interiors**, ultracompact configurations with trapped modes [6], and quantum backreaction or instabilities.

References

- [1] del Río & López-Oliver, *Spontaneous particle creation by oscillating compact stars*, Accepted for **Phys. Rev. D**, DOI: 10.1103/9s47-zspb [arXiv:2602.20253] (2026); [2] Birrell & Davies, *Quantum Fields in Curved Space* (1984); [3] Kokkotas & Ruoff, *Astron. Astrophys.* **366**, 565 (2001); [4] Fulling & Davies, *Proc. R. Soc. Lond. A* **348**, 393 (1976); [5] Wald, *Phys. Rev. D* **13**, 3176 (1976); [6] Cardoso & Pani, *Living Rev. Relativ.* **22**, 4 (2019).